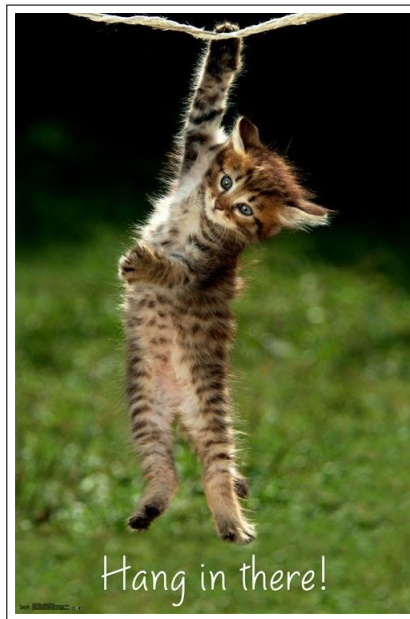


MTH310/MTH520: Graph Theory (GT)

Endsem Exam

May 4, 2025



Another motivational cat that gives you emotional support for a hundred and fifty minutes

Solve all 5 infant level questions.

For MTH310 students, you have the following two gameplay options:

- a) Solve any 4 out of 6 kiddo level questions.
- b) Solve any 2 out of 6 kiddo level questions and any 1 out of 3 adult level questions.

For MTH520 students, you have the following two gameplay options:

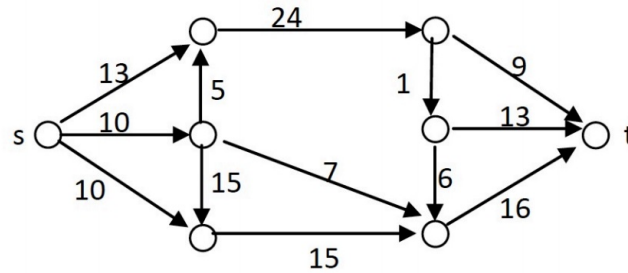
- a) Solve any 2 out of 6 kiddo level questions and any 1 out of 3 adult level questions.
- b) Solve any 2 out of 3 adult level questions.

If additional questions are solved, then the combination (according to the above forms) with the maximum possible score out of 30 marks will be selected.

Additional time and hints will be provided as per discretion.

§1 Infant Level

1. Compute the number of labelled spanning trees of K_n that contain n as a leaf. [CO2]
2. Show that every graph G has arboricity at least $\frac{d(G)}{4}$. [CO2]
3. Prove that $\forall p, q \geq 2$, $K_{p,q}$ is self-centered. [CO4]
4. Prove or disprove: There exists a bipartite planar connected graph G with $\delta(G) \geq 4$. [CO1]
5. Prove or disprove: In the given network, the maximum s - t flow is 26. [CO3]



[2+2+2+2+2=10 marks]

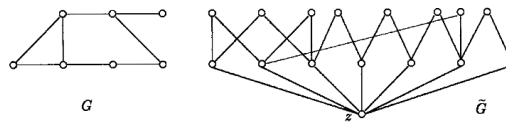
§2 Kiddo Level

1. [CO2, CO4] You have seen triangle-free graphs in Quiz 1 and median graphs in Quiz 2. Here is a construction which shows that they are actually closely related to each other:

Consider the following construction of $\tilde{G}$ from G :

- $V(\tilde{G}) := V(G) \cup E(G) \cup \{z\}$, where $z \notin V(G) \cup E(G)$
- $E(\tilde{G}) := \{\{z, v\} \mid v \in V(G)\} \cup \{\{v, e\} \mid v \in V(G), e \in E(G), v \in e \neq \phi\}$

An example of the construction is as follows:

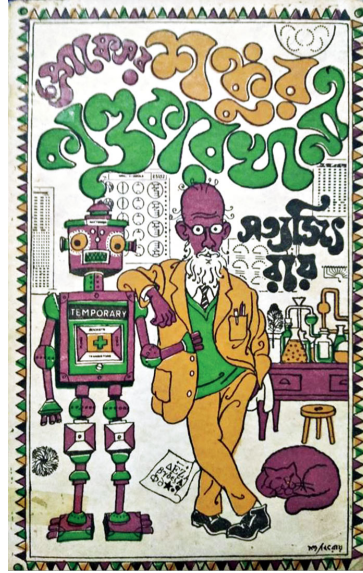


- a) Prove that $Q_3^- := Q_3 \setminus \{(0, 0, 0)\}$ is not a median graph, by showing that there exists a triplet of vertices with no possible median.
- b) A subgraph H of a graph G is called convex, if every shortest path between any two vertices in H , is contained within H itself. Show that Q_3^- is not a convex subgraph of Q_3 .
- c) Prove that if G is a median graph, G cannot have Q_3^- as a convex subgraph.
- d) Using part c) show that if $\tilde{G}$ is a median graph, then G is necessarily triangle-free. You may run into analyzing a few cases.

[1+1+0.5+2.5=5 marks]

2. [CO3, CO4] The crazy scientist Professor Trilokeshwar Shonku (প্রফেসর ত্রিলোকেশ্বর শঙ্কু) has discovered a new kind of particle which he denoted as an yugnion (যুগ্মায়ন), after some of them mysteriously appeared in his lab after an experiment. Some pairs of yugnions in the lab can be entangled, and each yugnion can participate in many entanglement relations with other yugnions. Shonku has found a way to perform the following two kinds of operations with these particles, one operation at a time.

- If some yugnion is entangled with an odd number of other yugnions in the lab, then Shonku can destroy it.
- At any moment, he may double the whole family of yugnions in his lab by creating a copy I' of each yugnion I . During this procedure, the two copies I' and J' become entangled if and only if the original yugnions I and J are entangled, and each copy I' becomes entangled with its original yugnion I ; no other entanglements occur or disappear at this moment.



Professor Shonku's Deeds (প্রফেসর শঙ্কুর কাণ্ডকারখানা) written by Satyajit Ray.

Prove that Shonku can apply a sequence of such operations resulting in a family of yugnions of no two of which are entangled.

(Hint: This question is about vertex colouring. How can the given operations be used to modify $\chi(G)$?)

[5 marks]

3. [CO3, CO4] Prove that every graph G with $\Delta(G) = \Delta$ has an equitable $\Delta + 1$ -edge colouring, which is defined as a $\Delta + 1$ -edge colouring where each colour is used either $\left\lfloor \frac{|E(G)|}{\Delta + 1} \right\rfloor$ and $\left\lceil \frac{|E(G)|}{\Delta + 1} \right\rceil$ many times in the colouring.

(Hint: This question is about matchings. What can you say about the union of two colour classes in an edge colouring?)

[5 marks]

4. [CO3] A question straight from the textbook (although paraphrased): Let $G = (X, Y, E)$ be a bipartite graph without isolated vertices, such that for every edge $\{x, y\} \in E$ where $x \in X, y \in Y$ we have $d(x) \geq d(y)$. Show that there exists an injective function $f : X \rightarrow Y$ such that $\{x, f(x)\} \in E \forall x \in X$.

[5 marks]

5. [CO4] Eskista (እስክስታ) is a traditional cultural dance performed by the Amhara ethnic group in Ethiopia. It is typically performed to traditional Amhara music, but it is possible to incorporate the style of dance into modern forms of music such as the music played in modern Ethiopian music videos. The complex nature of eskista dancing is what makes it arguably one of the most technical forms of African traditional dance.



Fendika (ፈንድካ) is an Ethiopian music group based in Addis Ababa, which was popularized in 2011 when Red Hot Chili Peppers released the song “Ethiopia” for their album “I’m with You” based on the band members’ time spent at Fendika.

In a performance with a set of dancers V where each dancer $v \in V$ mutually favours dancing with $d(v)$ many other dancers in V , show that the organizer can always choose at least $\sum_{v \in V} \frac{2}{1 + d(v)}$ many dancers, so that there is no cycle of dancers formed during the dance where each consecutive pair of dancers prefer each other as partners.

(Hint: Recall the probabilistic method that was used to show that every graph $G = (V, E)$ has an independent set of order at least $\sum_{v \in V} \frac{1}{1 + d(v)}$ in Assignment 1.)

[5 marks]

6. [CO3, CO4] A question straight outside of the textbook: Let G, H be a pair of graphs. Show that $\kappa(G \square H) \geq \kappa(G) + \kappa(H)$.

(Hint: Prove that $\kappa(G \square H) \geq \min\{\kappa(G) + \delta(H), \delta(G) + \kappa(H)\}$. What can you say from this?)

[5 marks]

§3 Adult Level

1. [CO4] As I had prophesized before, here are some questions based on dominating sets (from the Algorithmic Graph Theory topics covered in Tuts 7-9), which may additionally require an undergraduate level understanding of calculus and probability:

- a) Design a simple randomised algorithm that when given a graph $G = (V, E)$ with $\delta(G) = \delta > 1$, it outputs a dominating set with probability $\geq 1 - \frac{1}{n^2}$.

(Hint: The algorithm is a single line: Select every vertex in V independently and identically, with a fixed probability p . You may need to use the inequality $e^x \geq 1 + x \forall x \in \mathbb{R}$ to compute the exact value of p .)

- b) Show that G necessarily contains a dominating set D with $|D| \leq \frac{n + n \ln(\delta + 1)}{\delta + 1}$.

(Hint: Use a modified version of the random experiment in the previous part, for a probabilistic method style proof.)

- c) Not a fan of algorithms? Prove the following statement instead:

Let k be a positive integer and G be a graph. A subset U of $V(G)$ is called a k -path-dominating if for every $v \in V(G)$, there exists a path of length $\leq k$ from v to some element of U . **For every graph G , the number of all k -path-dominating subsets of G is always odd.**

[3+5+2=10 marks]

2. [CO1, CO4] A famous theorem that was mentioned in passing during the tutorials was the Hoffman-Singleton Theorem: If G is a d -regular graph ($d \geq 2$) with $g(G) \geq 5$ and $|G| = 1 + d^2 \implies d \in \{2, 3, 7, 57\}$. We will now use linear algebra to prove such a deep result in graph theory:

- a) If $A = A(G)$, give an explicit formula for $A + A^2$ in terms of the identity matrix I_n and all ones matrix J_n .

(Hint: Recall that $(A^k)_{i,j}$ is the number of i - j walks of length k .)

- b) Show that d is an eigenvalue of A . What is its corresponding eigenvector v ?
- c) Since A is real and symmetric, we can use the Spectral Theorem to extend $\{v\}$ to an orthogonal eigenvector basis of $\mathbb{R}^n$. Show that the multiplicity of the eigenvalue d is one, and A can have at most 2 other distinct eigenvalues apart from d .
- d) If $4d - 3$ is not a square, show that $d = 2$.

(Hint: Recall that $\text{Tr}(A)$ is the sum of all eigenvalues.)

- e) Let $s^2 = 4d - 3$, show that $s \mid 15$. Conclude the proof from here onwards.

(Hint: Recall that the product of all possible real roots of a polynomial $p(x) = a_k x^k + \dots + a_0$ of degree k is exactly $(-1)^k \frac{a_0}{a_k}$. If you add the extra condition that $s \in \mathbb{Z}$, what can you say now?)

[1.5+1+2.5+2+3=10 marks]

3. [CO3, CO4] Dr. Vadim Georgievich Vizing (Вадим Георгійович Візінг) was a Soviet mathematician best known for his groundbreaking work in graph theory, particularly in the study of graph colourings. The famous result known as Vizing's theorem was published by Vizing, when he was a PhD student in 1964. Instead of looking at generic graph colourings like Vizing did, we now focus on graph colourings that are unique (upto isomorphism):
- A graph G is said to be uniquely k -edge colourable if all k -edge colourings induce the same partition over $E(G)$. Let the partition for a k -edge colouring f be $M_i = \{e \in E(G) \mid f(e) = i\} \forall 1 \leq i \leq k$. Show that $G[M_i \cup M_j]$ has a Hamiltonian path (a path that visits all vertices) $\forall 1 \leq i < j \leq k$.
 - Let G be a cubic (3-regular) graph with a triangle T (3 edges with every pair adjacent on a common vertex). Show that G is uniquely 3-edge colourable if and only if $H = G \setminus T$ is also uniquely 3-edge colourable.
 - Are all uniquely 3-edge colourable graphs also 1-factorable? Prove this claim or give a counterexample.
 - A graph G is said to be uniquely k -colourable if all k -colourings induce the same partition over $V(G)$. Prove that $\chi(G) \leq \delta(G) + 1$.



Dr. Vadim Vizing (on the left) and Dr. Vitaly Voloshin (on the right) in Odessa, 2001. Dr. Vizing unfortunately passed away on 23 August 2017.

[3+2.5+1.5+3=10 marks]